Comparative Bayesian and Frequentist Volatility Modeling for Equities

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Abstract

This paper compares a GARCH(1,1) model with a Bayesian stochastic volatility (SV) model estimated via Hamiltonian Monte Carlo (HMC/NUTS) for modeling equity returns. Using daily data across both broad indices and highly volatile equities, we evaluate performance along two dimensions: the ability to reproduce empirical return characteristics through Monte Carlo simulation, and the accuracy of sequential predictive distributions.

The SV model produces more consistent higher-order distributional behavior and achieves higher log predictive likelihoods across the majority of assets considered. However, due to its centered volatility specification, it tends to underestimate variance in highly volatile equities, resulting in undercoverage of predictive intervals. The GARCH model, while less consistent in simulation and more sensitive to extreme observations, more accurately tracks realized volatility in high-variance regimes and produces yields higher empirical coverage of predictive intervals.

These results highlight a practical tradeoff: stochastic volatility models provide improved probabilistic fit and stability in moderate regimes, while GARCH-type models remain competitive in environments characterized by large, abrupt shocks.

1 Introduction

Modeling and forecasting financial time series remains a central problem in quantitative finance due to the presence of volatility clustering, heavy tails, and time-varying risk. A wide range of models have been developed to capture these features, with Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models forming one of the most widely used classical approaches. These models provide a tractable framework for conditional variance dynamics, but rely on deterministic updates that may limit their flexibility in complex or rapidly changing environments.

An alternative approach is provided by stochastic volatility (SV) models, which treat volatility as a latent stochastic process. In a Bayesian setting, these models naturally incorporate parameter uncertainty and allow for more flexible distributional assumptions. Recent advances in computational methods, particularly Hamiltonian Monte Carlo (HMC) and the No-U-Turn Sampler (NUTS), have made it feasible to estimate such models in practice, even with high-dimensional latent structures.

This paper compares a GARCH(1,1) specification with a Bayesian stochastic volatility model estimated via HMC/NUTS. The goal is to evaluate how differences in model structure – deterministic versus stochastic volatility dynamics – affect both simulation behavior and predictive performance across assets with varying levels of volatility.

To this end, we study a set of major equity indices and highly volatile individual stocks using approximately ten years of daily data. Model performance is evaluated along two dimensions: (i) the ability to reproduce key empirical features of returns through Monte Carlo simulation, and (ii) the quality of sequential predictive distributions generated in a rolling forecasting framework.

By examining both stable and highly volatile assets, this work aims to clarify when additional model flexibility leads to meaningful improvements, and when simpler parametric approaches remain sufficient.

2 Data and Problem Setup

This study focuses on modeling and forecasting the return dynamics of major U.S. equity indices, with some stable and some volatile. The full list is: the S&P 500 (SPY), T. Rowe S&P 500 Index Fund (PREIX), NASDAQ-100 (QQQ), Dow Jones Industrial Average (DIA), the Russell 2000 (IWM), Tesla (TSLA), Palantir (PLTR), Coinbase (COIN), Moderna (MRNA), and GameStop (GME). For each, we obtain daily price data spanning approximately ten years via the `yfinance` API, providing a sufficiently long horizon to capture multiple market regimes, including periods of elevated volatility.

From the raw price series, both simple returns and log returns are constructed. Simple returns are defined as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}},$$

while log returns are given by

$$\tilde{r}_t = \log\left(\frac{P_t}{P_{t-1}}\right).$$

For numerical stability during estimation, log returns are scaled by a factor of 100, a common practice in volatility modeling that improves the geometry of optimization and sampling algorithms without altering the underlying dynamics.

The primary objective of this work is twofold. First, we aim to evaluate each model's ability to generate realistic synthetic price paths via Monte Carlo simulation. A model is considered successful in this context if its simulated returns reproduce key empirical properties of financial time series, including mean behavior, volatility, tail risk, and autocorrelation structure.

Second, we assess predictive performance in a sequential forecasting setting. Specifically, models are trained on a historical window and evaluated on a held-out test period consisting of the most recent 252 trading days. At each step, the model produces a full predictive distribution for the next return, which is then compared against the realized outcome. This framework allows for the evaluation of both calibration and sharpness of the predictive distributions.

To quantify performance, we employ a range of statistical metrics. For simulation-based evaluation, we compare observed and simulated moments, distributional characteristics, and autocorrelation functions. For predictive evaluation, we measure empirical coverage of prediction intervals, average interval width, log predictive likelihood, and the distribution of probability integral transform (PIT) values.

This dual focus on simulation fidelity and predictive accuracy enables a comprehensive comparison between classical parametric models and more flexible Bayesian approaches, highlighting the tradeoffs between model simplicity, expressiveness, and practical performance.

3 Model Construction

3.1 GARCH Framework

To model conditional heteroskedasticity in asset returns, we employ a GARCH(1,1) specification with Student-t innovations. Let r_t denote the (scaled) log return at time t. The model is given by

$$r_t = \mu + \epsilon_t, \tag{1}$$

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu(0,1), \tag{2}$$

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{3}$$

where μ is the conditional mean, σ_t^2 is the conditional variance, and z_t follows a standardized Student-t distribution with ν degrees of freedom. The use of heavy-tailed innovations allows the model to better capture extreme return realizations commonly observed in financial markets.

Parameters $(\mu, \omega, \alpha, \beta, \nu)$ are estimated via maximum likelihood using the `arch` library. To ensure numerical stability and stationarity of the variance process, we impose the standard constraint

$$\alpha + \beta < 1.$$

In practice, if the estimated parameters approach the integrated GARCH (IGARCH) boundary, we rescale (α, β) such that $\alpha + \beta \leq 0.97$, preventing near-unit-root behavior and excessive persistence in the volatility process.

Overall, this framework provides a computationally efficient and interpretable benchmark for volatility modeling, while the inclusion of heavy-tailed innovations enhances its ability to capture key empirical features of financial time series.

3.2 Bayesian Stochastic Volatility (HMC/NUTS)

To provide a flexible alternative to parametric volatility models, we consider a Bayesian stochastic volatility (SV) specification in which log-volatility evolves as a latent stochastic process. Let r_t denote the (scaled) log return at time t. The model is defined as

$$r_t = \mu + \sigma_t z_t, \quad z_t \sim t_\nu(0, 1), \tag{4}$$

$$\sigma_t = \exp\left(\frac{h_t}{2}\right), \tag{5}$$

$$h_t = \phi h_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \tag{6}$$

where h_t is the latent log-volatility process, ϕ governs persistence, and σ_η controls the volatility of volatility. The use of Student-t innovations allows for heavy-tailed return behavior, consistent with empirical financial data. For this implementation, the latent log-volatility process is specified without an additional intercept term, centering it around zero. This improves identifiability and sampling stability in the HMC/NUTS framework, though at the cost of reduced flexibility in modeling long-run volatility levels.

Prior Specification. We adopt weakly informative priors to allow flexibility while maintaining numerical stability:

$$\mu \sim \mathcal{N}(0, 1), \tag{7}$$

$$\phi = 2u - 1, \quad u \sim \text{Beta}(1, 1), \tag{8}$$

$$\sigma_\eta \sim \text{HalfNormal}(1.5), \tag{9}$$

$$\nu = \nu_{\text{raw}} + 2, \quad \nu_{\text{raw}} \sim \text{Exponential}(1/10). \tag{10}$$

This parameterization ensures $\phi \in (-1, 1)$ for stationarity of the latent process and $\nu > 2$ so that the return variance is finite. For more volatile equities, one change was made to these priors:

$$\sigma_\eta \sim \text{HalfNormal}(0.2). \tag{11}$$

This alteration served to help mitigate funnel-like posterior geometry issues in HMC sampling.

Posterior Inference via HMC/NUTS. Inference is performed using Hamiltonian Monte Carlo (HMC) with the No-U-Turn Sampler (NUTS), as implemented in `PyMC`. This approach efficiently explores the high-dimensional posterior distribution over both parameters and latent states $\{h_t\}_{t=1}^T$. Compared to traditional MCMC methods, HMC leverages gradient information to produce lower autocorrelation in the sampled chains and improved convergence properties.

We generate multiple chains with a target acceptance probability of 0.95 to ensure stable exploration of the posterior geometry, particularly given the strong temporal dependence induced by the latent AR(1) structure.

4 Simulation Methodology

To evaluate the ability of each model to reproduce realistic financial dynamics, we generate forward price paths via Monte Carlo simulation. While both models produce predictive distributions, their simulation mechanisms differ fundamentally due to their underlying structure.

4.1 GARCH Simulation

Simulation under the GARCH framework proceeds via forward recursion of the conditional variance equation using estimated parameters from the fitted model. Let $(\mu, \omega, \alpha, \beta, \nu)$ denote the maximum likelihood estimates, and let (σ_T, ϵ_T) be the final in-sample volatility and residual.

At each step of the simulation horizon, the conditional variance is updated according to

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \tag{12}$$

A standardized Student-t innovation is then drawn,

$$z_t \sim t_\nu(0, 1), \tag{13}$$

and rescaled to ensure unit variance:

$$z_t \leftarrow z_t \cdot \sqrt{\frac{\nu - 2}{\nu}}.$$

Returns are generated as

$$r_t = \mu + \sigma_t z_t, \tag{14}$$

Stability Adjustments. To prevent explosive variance behavior, we continue to constrain the persistence parameters such that $\alpha+\beta \leq 0.97$. This avoids near-integrated GARCH dynamics during forward simulation.

Price Construction. Simulated log returns are accumulated and exponentiated to obtain price paths:

$$P_t = P_0 \exp\left(\sum_{i=1}^{t} r_i\right).$$

where P_0 is the most recent observed price. Since returns are modeled in log space, this transformation ensures consistency with continuous compounding.

4.2 Bayesian Stochastic Volatility Simulation

Simulation under the Bayesian stochastic volatility framework proceeds by drawing from the posterior distribution over model parameters and propagating the latent volatility process forward.

Posterior Sampling. For each simulated path, a parameter tuple $(\mu, \phi, \sigma_\eta, \nu)$ is randomly drawn from the posterior samples obtained via HMC/NUTS. This ensures that parameter uncertainty is fully incorporated into the simulation.

Latent Volatility Evolution. The latent log-volatility process is evolved recursively as

$$h_t = \phi h_{t-1} + \sigma_\eta \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1). \tag{15}$$

Return Generation. At each time step, returns are generated using

$$r_t = \mu + \exp\left(\frac{h_t}{2}\right) z_t, \quad z_t \sim t_\nu(0, 1), \tag{16}$$

Price Construction. As in the GARCH case, simulated returns are converted into price paths via exponential compounding:

$$P_t = P_0 \exp\left(\sum_{i=1}^{t} r_i\right).$$

As in the GARCH case, this ensures consistency with log-return dynamics.

5 Sequential Predictive Evaluation

5.1 Particle Filtering for HMC

To evaluate the predictive performance of the Bayesian stochastic volatility model, we implement a sequential posterior predictive framework using particle filtering. The goal is to propagate uncertainty in both the latent volatility process and the model parameters forward through time, producing a full predictive distribution for each subsequent return.

Particle Representation. Each particle represents a sampled parameter tuple and a current latent volatility state (of N particles):

$$(\phi, \sigma_\eta, \mu, \nu, h_t)^{(i)}, \quad i = 1, \ldots, N.$$

These particles are drawn with replacement from the posterior samples obtained via HMC/NUTS, ensuring that both parameter uncertainty and variability in the last latent state are incorporated into the sequential prediction.

Sequential Predict Step. At each time step, particles are propagated forward according to the latent volatility dynamics:

$$h_{t+1}^{(i)} = \phi^{(i)} h_t^{(i)} + \sigma_\eta^{(i)} \epsilon_t^{(i)}, \quad \epsilon_t^{(i)} \sim \mathcal{N}(0, 1),$$

Conditional returns are then generated as

$$r_{t+1}^{(i)} = \mu^{(i)} + \exp\left(\frac{h_{t+1}^{(i)}}{2}\right) z_{t+1}^{(i)}, \quad z_{t+1}^{(i)} \sim t_{\nu^{(i)}}(0, 1),$$

producing a predictive distribution for the next observed return.

Weighting and Resampling. Observed returns are used to compute particle weights via the predictive likelihood:

$$w_{t+1}^{(i)} \propto p\big(r_{t+1}^{\text{obs}} \mid h_{t+1}^{(i)}, \theta^{(i)}\big)$$

where $\theta = (\mu, \phi, \sigma_\eta, \nu)$ denotes the vector of model parameters. Particles are then resampled according to these weights, which focuses computational effort on trajectories consistent with observed data while maintaining diversity across the posterior parameter space. This resampling step is repeated sequentially for each return in the test set.

By combining posterior draws with sequential propagation and likelihood-based resampling, this particle filtering approach provides a principled way to generate predictive distributions that account for both latent volatility dynamics and parameter uncertainty. It also enables subsequent evaluation via interval coverage, probability integral transform (PIT) calibration, and log predictive likelihood, while maintaining a fully probabilistic interpretation of the forecasts.

5.2 Rolling Forecast for GARCH

For the classical GARCH(1,1) model, sequential predictive evaluation is performed using a rolling forecast framework. Unlike the particle filtering approach for the Bayesian model, the GARCH predictive distribution is generated directly from the model's conditional variance recursion.

Conditional Variance Propagation. At each time step, the one-step-ahead conditional variance is updated according to the standard GARCH(1,1) recursion:

$$\sigma_{t+1}^2 = \omega + \alpha\, r_t^2 + \beta\, \sigma_t^2,$$

where r_t is the most recently observed return and σ_t^2 is the previous conditional variance. This recursion incorporates both recent return shocks and the persistence of past volatility.

Predictive Distribution. Given the conditional variance σ_{t+1}^2 and the estimated GARCH parameters $(\mu, \alpha, \beta, \omega, \nu)$, the predictive distribution for the next return is assumed to follow a Student-t law:

$$r_{t+1} \sim \mu + \sigma_{t+1}\, z, \quad z \sim t_\nu(0,1),$$

with the Student-t standardized to unit variance. This allows the model to capture heavy-tailed return behavior while maintaining the time-varying volatility structure implied by the GARCH recursion.

Sequential Update. After observing the actual return r_{t+1}^{obs}, the residual is updated:

$$r_{t+1}^{\text{resid}} = r_{t+1}^{\text{obs}} - \mu,$$

which is then used to propagate the conditional variance for the subsequent time step. This rolling update produces a sequential predictive distribution for each point in the test set.

The rolling forecast framework leverages the autoregressive structure of conditional variance in GARCH models to sequentially generate predictive distributions. While it does not explicitly propagate parameter uncertainty as in the particle filtering method, it provides a straightforward and computationally efficient mechanism to assess predictive performance through interval coverage, probability integral transform calibration, and log predictive likelihood.

6 Simulation-Based Metrics

We begin by evaluating each model's ability to reproduce key statistical properties of observed return series through Monte Carlo simulation. The focus is on unconditional moments, tail behavior, and volatility clustering, which together characterize the empirical distribution of financial returns. Further, we evaluate divergences in the HMC and GARCH models' performances and how they reflect upon the models' inherent structures.

6.1 Stable Equities

First and Second Moments. Across all indices, both models successfully reproduce the mean of returns, with simulated values remaining close to observed averages. However, systematic differences emerge in volatility estimation. For stable equities, the Bayesian stochastic volatility (SV) model consistently produces standard deviations that are slightly higher than observed values, while the GARCH model tends to produce bounds that are too tight to accurately reflect the volatile periods of this dataset. This pattern is consistent across datasets, suggesting that the SV model captures smoother volatility dynamics, while GARCH exhibits stronger mean-reversion that dampens variability in this setting.

	SPY	QQQ	DIA	IWM	PREIX
Observed Mean	0.000588	0.000794	0.000523	0.000489	0.000598
GARCH Simulated Mean	0.001052	0.001465	0.000904	0.000731	0.001033
HMC Simulated Mean	0.001148	0.001572	0.000863	0.000805	0.001146

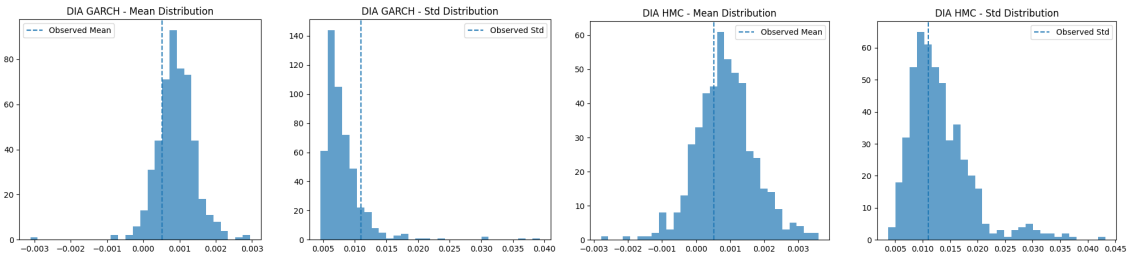
Distribution of Observed/Simulated Means

	SPY	QQQ	DIA	IWM	PREIX
Observed SD	0.011270	0.013993	0.011011	0.014468	0.011350
GARCH Simulated SD	0.007987	0.009181	0.008184	0.010146	0.007993
HMC Simulated SD	0.013527	0.012679	0.013185	0.011558	0.013235

Distribution of Observed/Simulated Std. Dev.

Tail Behavior and Kurtosis. A key distinction between the models arises in their ability to capture heavy tails. Empirical return distributions exhibit substantial excess kurtosis (e.g., values exceeding 15 in several indices), reflecting the presence of extreme events.

For the most part, both the GARCH and the SV model produce kurtosis levels close to observed data, occasionally slightly under- or over-estimate depending on the asset. The SV model is slightly more consistent, however, while the GARCH model exhibits extreme behavior in higher moments, occasionally producing extreme kurtosis values. We should note that these extreme cases are more significant when we calculate pooled kurtosis. When we recalculate the per-path average of local kurtosis, GARCH performs more reasonably as the occasional extreme paths fail to skew the data as significantly. This reflects the sensitivity of GARCH to large residual shocks, which can propagate disproportionately through the variance recursion if the resulting data is not effectively monitored.



Ex: DIA Index

Tail Risk Measures. We further assess tail behavior using Value-at-Risk (VaR) and Expected Shortfall (ES). The SV model produces tail quantiles that are generally more consistent with observed data, while GARCH tends to either underestimate moderate tail risk or generate excessively extreme outcomes in rare cases.

This asymmetry reflects the structural difference between the models: GARCH deterministically updates volatility based on past squared returns, whereas the SV model allows volatility itself to evolve stochastically, leading to a smoother but more flexible tail structure.

Volatility Clustering. To evaluate temporal dependence in volatility, we compare autocorrelation functions (ACF) of squared returns. Both models capture the persistence of volatility clustering, though the mechanisms differ. GARCH directly encodes persistence through the $\alpha + \beta$ parameter, while the SV model captures it through the autoregressive coefficient ϕ in the latent volatility process.

Empirically, the SV model produces smoother and more gradually decaying autocorrelation profiles, whereas GARCH often exhibits sharper short-term persistence followed by faster decay. Models were similarly effective and average ACF was reasonable across HMC and GARCH.

6.2 Volatile Equities

We now turn to assets characterized by substantially higher volatility, including TSLA, PLTR, COIN, MRNA, and GME. These securities exhibit markedly different statistical properties compared to broad equity indices, including elevated variance, more frequent extreme returns, and in some cases pronounced regime shifts driven by idiosyncratic factors.

First and Second Moments. A clear divergence between the models emerges in their ability to reproduce observed volatility levels. Across all volatile equities, the Bayesian stochastic volatility (SV) model significantly underestimates return variance. For example, in TSLA, the observed standard deviation is approximately 0.0372, while the SV model produces a simulated value near 0.0118. Similar patterns appear for PLTR (0.0442 vs 0.0114), COIN (0.0542 vs 0.0115), and MRNA (0.0456 vs 0.0112).

In contrast, the GARCH model more closely tracks observed volatility in most cases. For instance, GARCH produces a standard deviation of 0.0434 for PLTR and 0.0506 for COIN, both closely aligned with

empirical values. However, it still exhibits underestimation in certain assets (e.g., TSLA), suggesting some limitations in capturing the full scale of volatility.

	TSLA	PLTR	COIN	MRNA	GME
Observed SD	0.037164	0.044224	0.054214	0.045552	0.074170
GARCH Simulated SD	0.017637	0.043360	0.050647	0.030721	0.069175
HMC Simulated SD	0.011780	0.011443	0.011487	0.011202	0.016260

Distribution of Observed/Simulated Std. Dev.

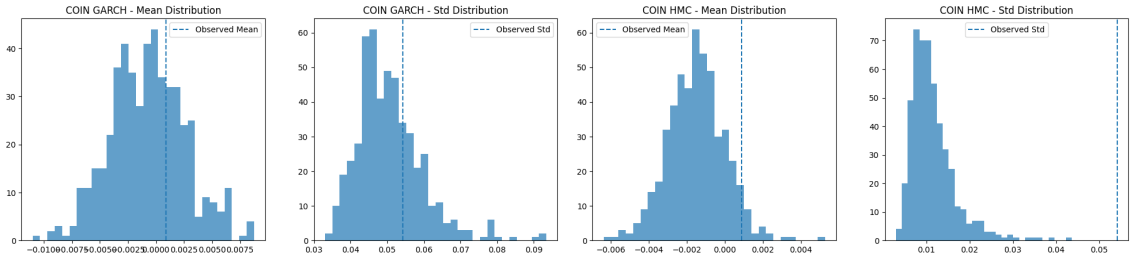
This discrepancy reflects a combination of model specification and prior-induced shrinkage in the stochastic volatility framework. In particular, the absence of an explicit long-run volatility level limits the model's ability to sustain elevated variance regimes.

Tail Behavior and Kurtosis. Volatile equities exhibit a wide range of tail behaviors, from moderate kurtosis (e.g., COIN at 2.73) to extreme heavy tails (e.g., GME at 83.85). The SV model generally produces stable kurtosis values, often overshooting moderate tails while remaining bounded in extreme cases. For instance, it generates a pooled kurtosis of 14.13 for TSLA and 14.32 for COIN, significantly higher than observed values. It should be noted that, even in these cases, the path-wise kurtosis of the Bayesian SV remains true to the dataset (much as GARCH was for stable equities), but overall its stability remains impaired compared to the previous setting. Altogether, this reflects the model's tendency to generate consistently heavy tails rather than sporadic extreme outliers.

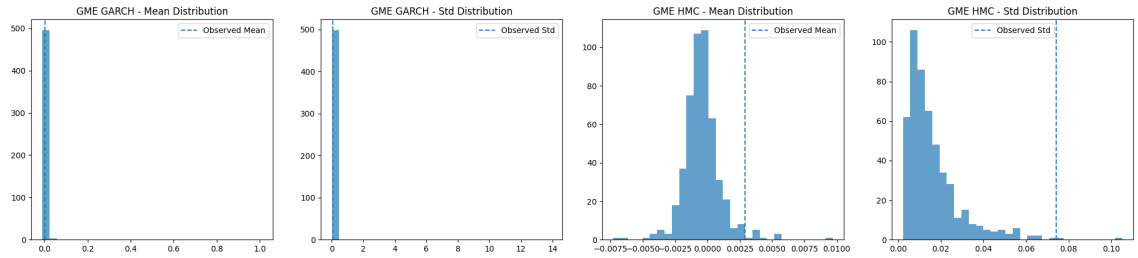
The GARCH model, by contrast, demonstrates sensitive tail behavior. For some assets, it drastically overestimates kurtosis (across both calculations) due to heavy-tailed behavior. Its path-wise kurtosis continues to produce reasonable estimates for some assets (e.g., COIN at 2.5187), it becomes numerically reactive in extreme cases.

It should be noted that both models struggled significantly to map the tails of GME due to the extreme but rare price swings that the index has experienced. Whereas other indices might be consistently volatile, consistently stable, or somewhere in between, this illustrates how the structural assumptions of both Bayesian and frequentist models can struggle when faced with outliers such as this.

Overall, this behavior highlights a fundamental tradeoff: the SV model produces smoother tail estimates, while GARCH can better match moderate tails but may perform poorly in the presence of extreme outliers.



Ex: COIN Equity



Ex: GME Equity

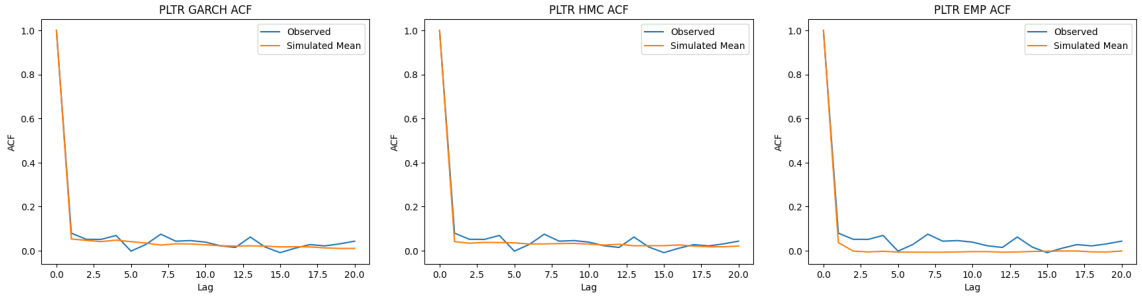
Tail Risk Measures. Consistent with the kurtosis results, the SV model tends to produce smoother but less extreme tail risk estimates. In highly volatile assets, this leads to underestimation of true risk exposure, as the model fails to generate sufficiently large deviations.

The GARCH model, on the other hand, often produces more realistic tail risk estimates for moderately volatile assets (e.g., PLTR, COIN), but may generate excessively extreme outcomes in assets with large shocks (e.g., GME). This results in a bimodal failure mode: either underestimating risk or dramatically overshooting it.

Volatility Clustering. Both models capture volatility clustering, though with distinct characteristics. The SV model produces smoother and more persistent volatility dynamics, reflecting the AR(1) structure of the latent log-volatility process. However, due to its variance underestimation, the magnitude of clustering is muted.

The GARCH model more accurately captures the magnitude of volatility bursts, particularly in assets such as COIN and GME, where large shocks are rapidly incorporated into the conditional variance. However, this responsiveness can lead to excessive persistence following extreme events, contributing to the instability observed in higher-order moments.

Once again, we aggregate values and compare ACF to judge fit. Average ACF remains reasonable for both HMC and GARCH. Once smoothed, data reflects relative agreement.



Ex: PLTR Equity (Right: Empirical MCMC baseline)

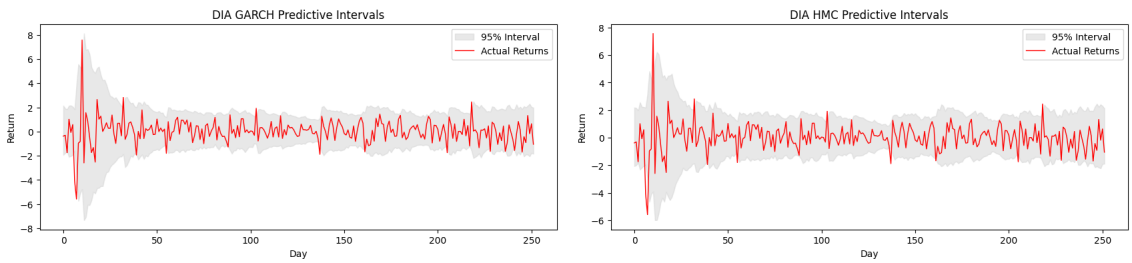
7 Predictive Performance

We now evaluate the sequential predictive performance of each model using a rolling forecasting framework. At each time step, models produce a full predictive distribution for the next return, which is evaluated against observed outcomes. Again, we compare distributions between HMC and GARCH models.

7.1 Stable Equities

Interval Coverage and Sharpness. Both models achieve empirical coverage rates close to the nominal 95% level across all indices. For stable indices, the SV model consistently exhibits slightly higher coverage (typically between 95.6% and 96.8%) compared to GARCH (approximately 94.0% to 96.0%).

This difference reflects a tradeoff between calibration and sharpness. The SV model produces wider predictive intervals, leading to mild overcoverage, while GARCH generates comparatively narrower intervals that are closer to nominal levels but occasionally under-cover. The increased width in the SV model arises from the propagation of both latent volatility uncertainty and parameter uncertainty, resulting in more diffuse predictive distributions. That being said, both approaches demonstrate well-attuned intervals and coverage.



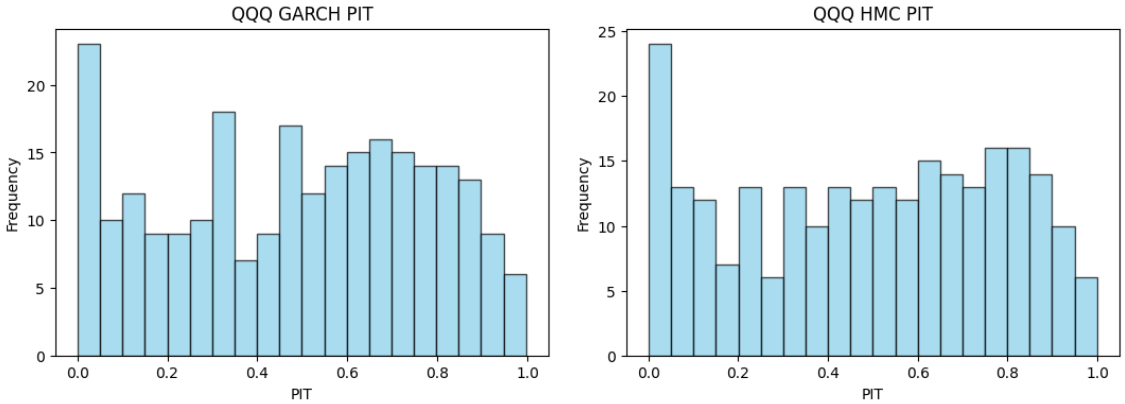
Ex: DIA Equity

Log Predictive Likelihood. The SV model consistently achieves higher (less negative) log predictive likelihoods across all assets, indicating a better fit to realized returns. This improvement reflects the model's ability to capture both parameter uncertainty and stochastic volatility dynamics.

The difference, while modest in magnitude, is consistent across datasets, suggesting that the SV framework provides a systematically better probabilistic description of return distributions.

PIT Calibration. Probability Integral Transform (PIT) values are used to assess calibration of predictive distributions. Both models produce PIT means close to 0.5, indicating unbiased predictions. However, differences arise in the dispersion of PIT values.

The SV model tends to produce slightly lower PIT variance, suggesting improved calibration and a closer match between predicted and realized distributions. GARCH, by contrast, exhibits slightly higher variance, consistent with occasional under- or over-confidence in its predictive intervals.



Ex: QQQ Index

Sequential Adaptation. A key distinction between the models lies in how they adapt to new information. The particle filtering approach used with the SV model dynamically updates both latent states and parameter weights, allowing the model to respond flexibly to changing volatility regimes.

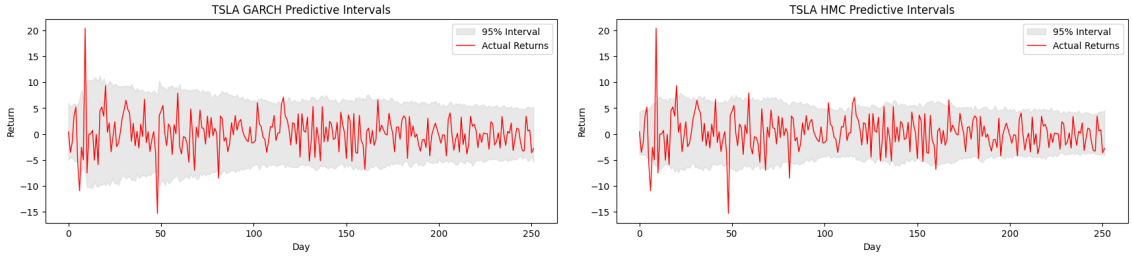
In contrast, the GARCH model updates only the conditional variance through a fixed parameter structure. While computationally efficient, this limits its ability to adapt to structural shifts in the data.

7.2 Volatile Equities

We now evaluate predictive performance for volatile equities, where large return magnitudes and rapid regime shifts place greater demands on model flexibility and calibration.

Interval Coverage and Sharpness. A notable reversal occurs in coverage behavior relative to stable equities. The SV model consistently undercovers, with empirical coverage rates ranging from approximately 88% to 92% across assets. For example, coverage is 0.9167 for TSLA, 0.8849 for PLTR, and 0.8929 for COIN. In contrast, the GARCH model generally achieves or exceeds nominal coverage levels, often producing conservative intervals. Coverage reaches 0.9683 for TSLA and 0.9762 for GME, indicating overcoverage in some cases.

This difference is directly tied to the variance estimation behavior observed in the simulation results. The SV model's underestimation of volatility leads to intervals that are too narrow, while GARCH's responsiveness to large shocks produces wider, more conservative intervals. Average interval widths further reinforce this pattern. For instance, in TSLA, the SV interval width is approximately 10.73, compared to 13.70 under GARCH. Similar differences are observed across all volatile assets.



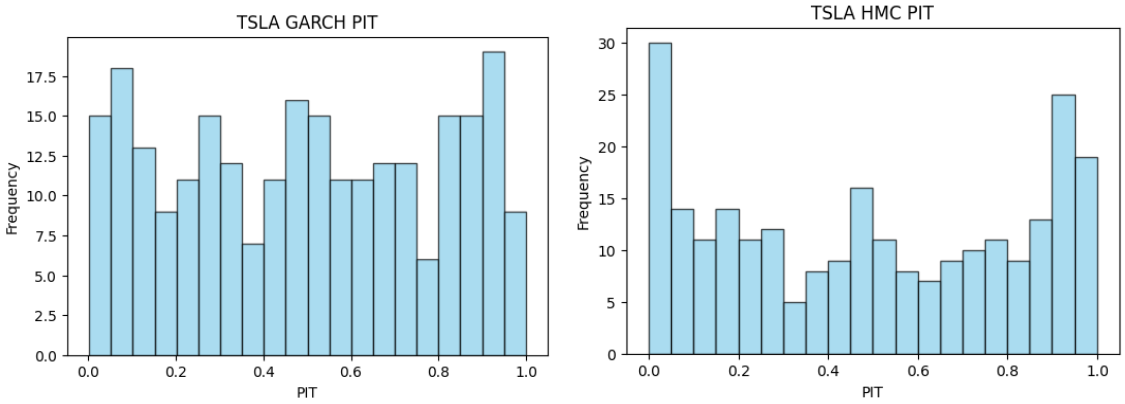
Ex: TSLA Equity

Log Predictive Likelihood. Despite its undercoverage, the SV model consistently achieves higher (less negative) log predictive likelihoods across all volatile equities. For example, the SV model attains a log likelihood of -2.596 for TSLA compared to -2.633 under GARCH, and -2.888 vs -2.938 for COIN.

This suggests that while SV underestimates extreme events, it provides a better overall fit to the bulk of the return distribution. In contrast, GARCH's heavier tails and wider intervals improve coverage but reduce sharpness, leading to slightly worse likelihood performance.

PIT Calibration. Both models maintain approximately unbiased PIT means near 0.5 across all assets. However, differences arise in variance. The SV model exhibits higher PIT variance (e.g., 0.1088 for TSLA, 0.1106 for PLTR), indicating underdispersion and insufficient uncertainty quantification.

The GARCH model produces slightly lower PIT variance (e.g., 0.0877 for TSLA, 0.0713 for COIN), consistent with confident predictive distributions. This aligns with its tendency toward conservative interval estimates for this dataset.



Ex: TSLA Equity

Sequential Adaptation. In volatile environments, the ability to rapidly adapt to new information is critical. The particle filtering framework used in the SV model allows for dynamic updating of latent volatility states and parameter weights. However, the effectiveness of this adaptation is limited by the model's structural inability to generate sufficiently large volatility levels.

The GARCH model, despite its fixed parameterization, responds more aggressively to large return shocks through its variance recursion. This enables it to better capture abrupt volatility spikes, particularly in assets such as COIN and GME.

Overall, the results for volatile equities highlight a reversal of the performance hierarchy observed in stable assets when it comes to predictive performance. While the SV model remains superior in likelihood-based metrics, its inability to capture high volatility regimes leads to systematic undercoverage and underestimation of risk.

The GARCH model, despite its instability in simulation and tendency toward overconservatism, provides more reliable predictive intervals in high-volatility settings. This suggests that, in practice, simpler parametric models may remain competitive—or even preferable—when modeling assets characterized by extreme variability and frequent large shocks.

8 Discussion

The results of this study highlight a fundamental tradeoff between structural flexibility and numerical stability in volatility modeling. While both the GARCH framework and the Bayesian stochastic volatility (SV) model are capable of reproducing key stylized facts of financial returns, their behavior diverges in systematic ways across both simulation and predictive settings.

Model Structure and Implications. The distinction between the models is primarily structural. The GARCH model updates conditional variance deterministically as a function of past squared residuals, which leads to fast adaptation to shocks but also introduces sensitivity to extreme observations. In contrast, the SV model treats volatility as a latent stochastic process, allowing for smoother transitions in volatility regimes and a natural incorporation of parameter uncertainty through posterior inference.

This difference explains several of the observed empirical patterns. The SV model produces more stable higher-order moments and more consistent tail behavior in simulation, while GARCH can generate extreme outcomes due to recursive amplification of large residuals. At the same time, the deterministic structure of GARCH enables it to more directly track realized volatility in high-variance environments.

Stable vs. Volatile Regimes. A key result is the divergence in model performance between stable indices and highly volatile equities. For broad indices, the SV model provides a better overall probabilistic

description of returns, as reflected in improved calibration and higher log predictive likelihoods. This is consistent with the relatively smooth evolution of volatility in such assets, which aligns well with the latent AR(1) structure of the SV specification.

However, in highly volatile equities, the SV model systematically underestimates variance. This behavior is directly attributable to the centered specification of the latent log-volatility process, which restricts the model's ability to sustain elevated volatility levels. In contrast, GARCH is able to more accurately match observed variance due to its direct dependence on realized shocks, though this comes at the cost of occasional instability. These results also suggest that model misspecification can dominate methodological differences, as the absence of a long-run volatility level in the SV model materially impacts performance in high-volatility regimes.

Predictive Tradeoffs. From a predictive standpoint, the SV model demonstrates consistently higher log predictive likelihoods, indicating a better fit to the bulk of the return distribution. However, this improvement does not uniformly translate into better risk estimation. In volatile assets, the SV model produces intervals that are too narrow, leading to systematic undercoverage.

GARCH exhibits the opposite behavior: predictive intervals are wider and often conservative, resulting in higher coverage but reduced sharpness. This reflects a broader tradeoff between calibration and responsiveness. The SV model provides a more coherent probabilistic framework, while GARCH offers more robust behavior in the presence of large shocks.

Adaptation and Computational Considerations. The particle filtering framework used with the SV model enables dynamic updating of both latent states and parameter weights, providing a principled mechanism for sequential learning. However, this flexibility is computationally expensive and sensitive to prior specification. In practice, achieving stable performance across assets of differing volatility required retuning of priors, particularly for the volatility-of-volatility parameter.

The GARCH model, by contrast, is computationally efficient and largely robust across datasets without modification. While less expressive, its simplicity makes it more practical for applications requiring scalability or real-time forecasting.

Overall, the results suggest that the advantages of Bayesian stochastic volatility models are most pronounced in environments where volatility evolves smoothly and parameter uncertainty is relevant. In contrast, for assets characterized by large, abrupt shocks, simpler parametric models such as GARCH may provide more reliable variance tracking despite their limitations.

9 Conclusion

This paper presents a comparative analysis of a GARCH(1,1) model and a Bayesian stochastic volatility model estimated via HMC/NUTS, evaluated across both simulation fidelity and sequential predictive performance.

The findings indicate that Bayesian stochastic volatility models offer a more flexible and internally consistent framework for modeling financial returns, particularly in terms of distributional accuracy and probabilistic calibration. The incorporation of latent volatility dynamics and parameter uncertainty leads to improved log predictive likelihoods and more stable higher-order moment behavior in simulation.

However, these advantages are conditional on appropriate model specification. In the absence of a long-run volatility level, the SV model systematically underestimates variance in high-volatility assets, leading to undercoverage and weaker performance in tail risk estimation. This highlights the importance of careful model design, particularly when applying Bayesian methods to heterogeneous datasets.

The GARCH model, despite its relative simplicity, remains competitive across a wide range of settings. Its ability to directly incorporate realized shocks allows it to better match observed volatility in extreme environments, and its computational efficiency makes it well-suited for practical implementation. At the same time, its sensitivity to large residuals can lead to instability in simulation and less consistent probabilistic calibration.

Taken together, these results reinforce the view that model selection in volatility modeling is inherently context-dependent. Bayesian approaches provide clear advantages in terms of flexibility and probabilistic coherence, while classical parametric models offer robustness and efficiency. In practice, the choice between these frameworks should be guided by the characteristics of the asset being modeled and the specific objectives of the analysis.

Future work may focus on extending the SV specification to include a mean-reverting volatility level or hierarchical structure, as well as exploring hybrid approaches that combine the strengths of both frameworks. Future work may also consider performance-oriented implementations of HMC in lower-level languages such as C++ or C to improve computational efficiency. Such extensions may help bridge the gap between flexibility and stability observed in this study.